1 Question (10 points)

Describe a multitape Turing machine with 5 tapes that recognizes the language $L = \{0^n 1^m 2^n 3^m | n, m \geq 1\}$. You need to use all tapes in your solution. Do not design the TM. Describe the steps it needs to take.

2 Question (15 points)

Design a TM that recognizes the language $L = \{0^n 1^m 2^n 3^m | n, m \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

3 Question (10 points)

Consider the language

$L = \{< D, k > | D$ is DFA and it rejects all strings w if $k < |w| < m\}$, where $k, m \in N$

Is L decidable? Prove it.

4 Question (15 points)

Describe a TM that recognizes the language $L = \{0^n1^m2^p3^{n+m+p} | n, m, p \geq 1\}$. Do not design the TM. Describe the steps it needs to take.